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AD TECH PRIMER

US Internet

Intro to Mobile Ad Tech: Who Are the Players, What Do They Do, and What's Changing?

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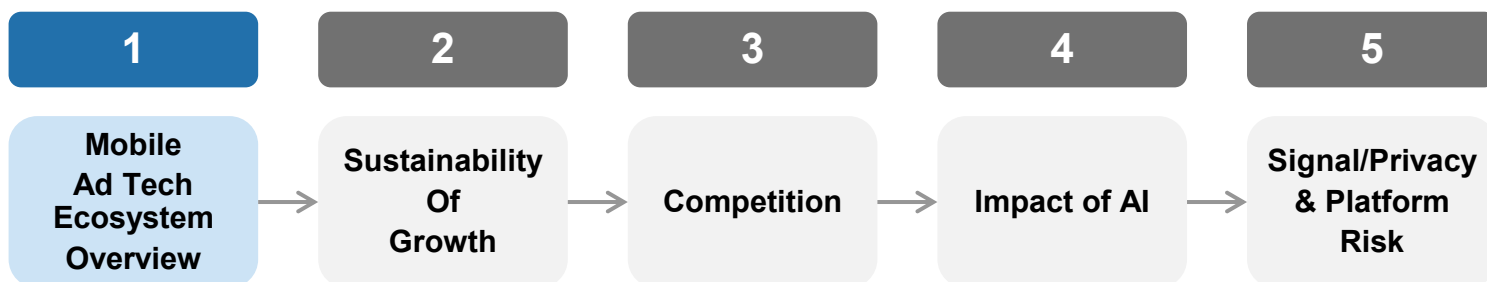
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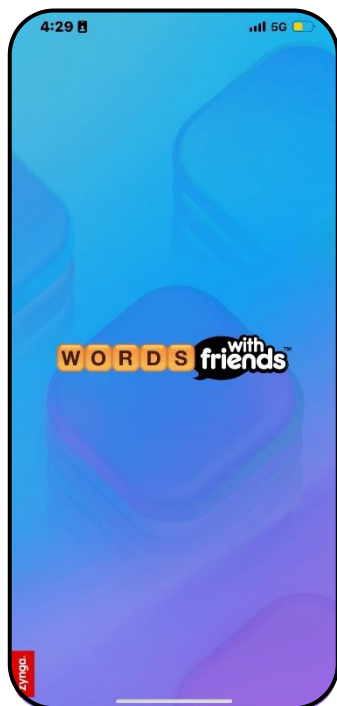
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Intro to Mobile Ad Tech



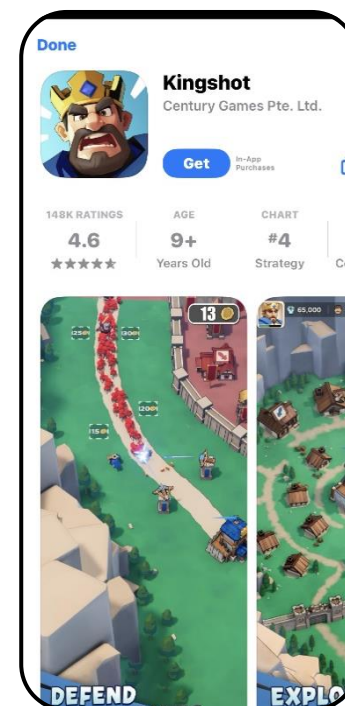
In-App Ads Typically Send Consumers to the App Stores to Download New Products – Today, Most of Those Ads Are Promoting Games...



User Opens & Plays a Publisher's App



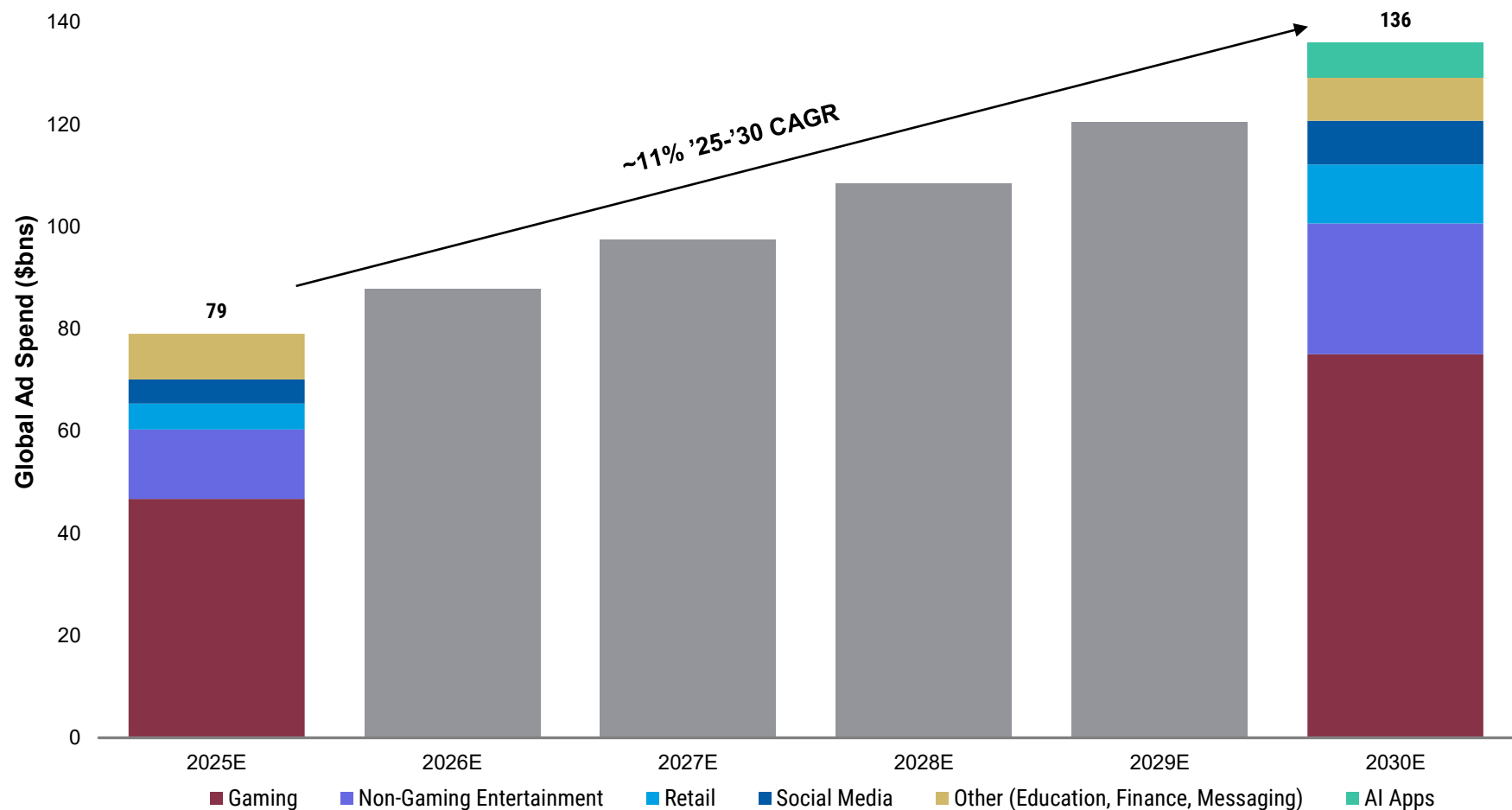
An Interactive or Video Ad Will Appear for User



User is Prompted to Download the Game

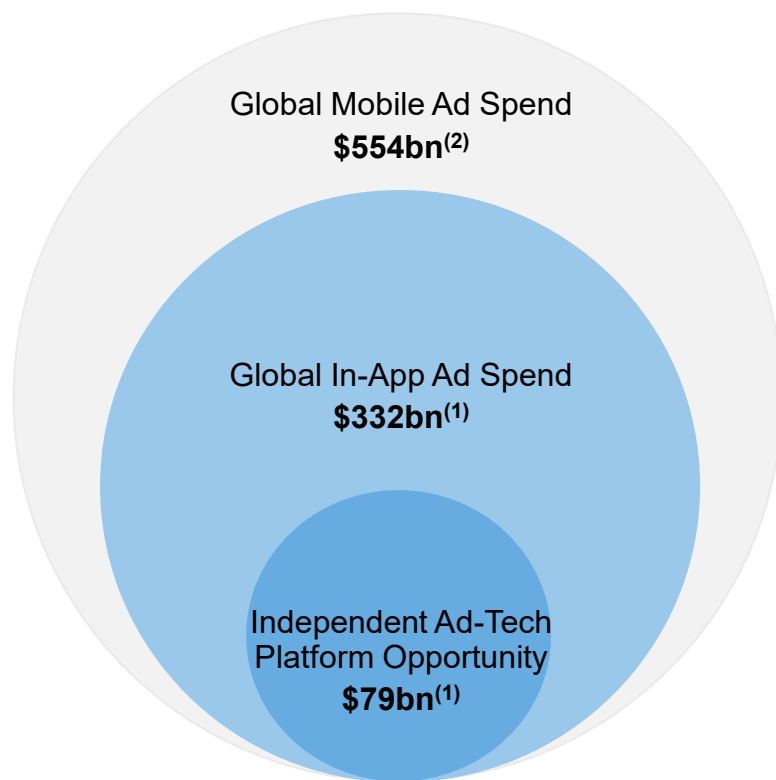
...But Other Categories like Retail and Entertainment Are Important Drivers of the \$79bn Opportunity for Independent In-App Ad Tech Platforms Like APP and U

Global Ad Spend on Independent In-App Ad Tech Platforms is Projected to Reach \$136bn in 2030*



*According to Altman Solon
Source: Company Data, Morgan Stanley Research, Altman Solon

The Estimated Near-Term Market Opportunity is \$79bn, Though Total In-App and Mobile Advertising Are Larger, When Including Walled Garden Platforms like GOOGL and META



5bn
Smartphone Users Globally⁽²⁾

~3 hours
Spent on Apps per Day⁽³⁾

\$554bn
Global Mobile Ad Spend per Year⁽²⁾

90%
Mobile Time Spent In-App⁽²⁾

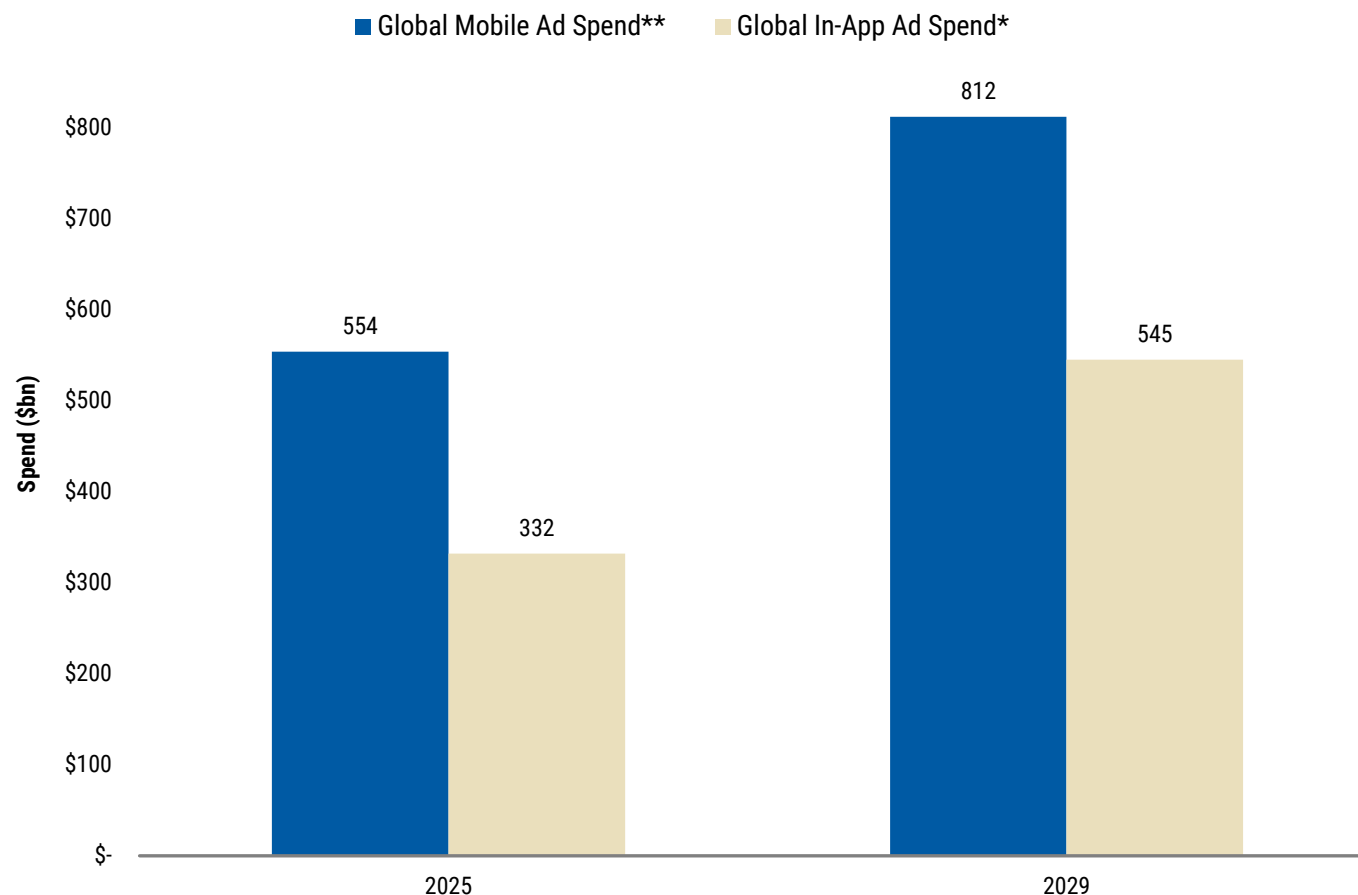
(1) According to Altman Solon

(2) According to eMarketer

(3) According to SensorTower

Source: Company Data, Morgan Stanley Research, Altman Solon, eMarketer, SensorTower

We Also Expect Meaningful Growth Across Broader Markets Like In-App Advertising (incl. Walled Gardens) as well as Total Mobile Advertising



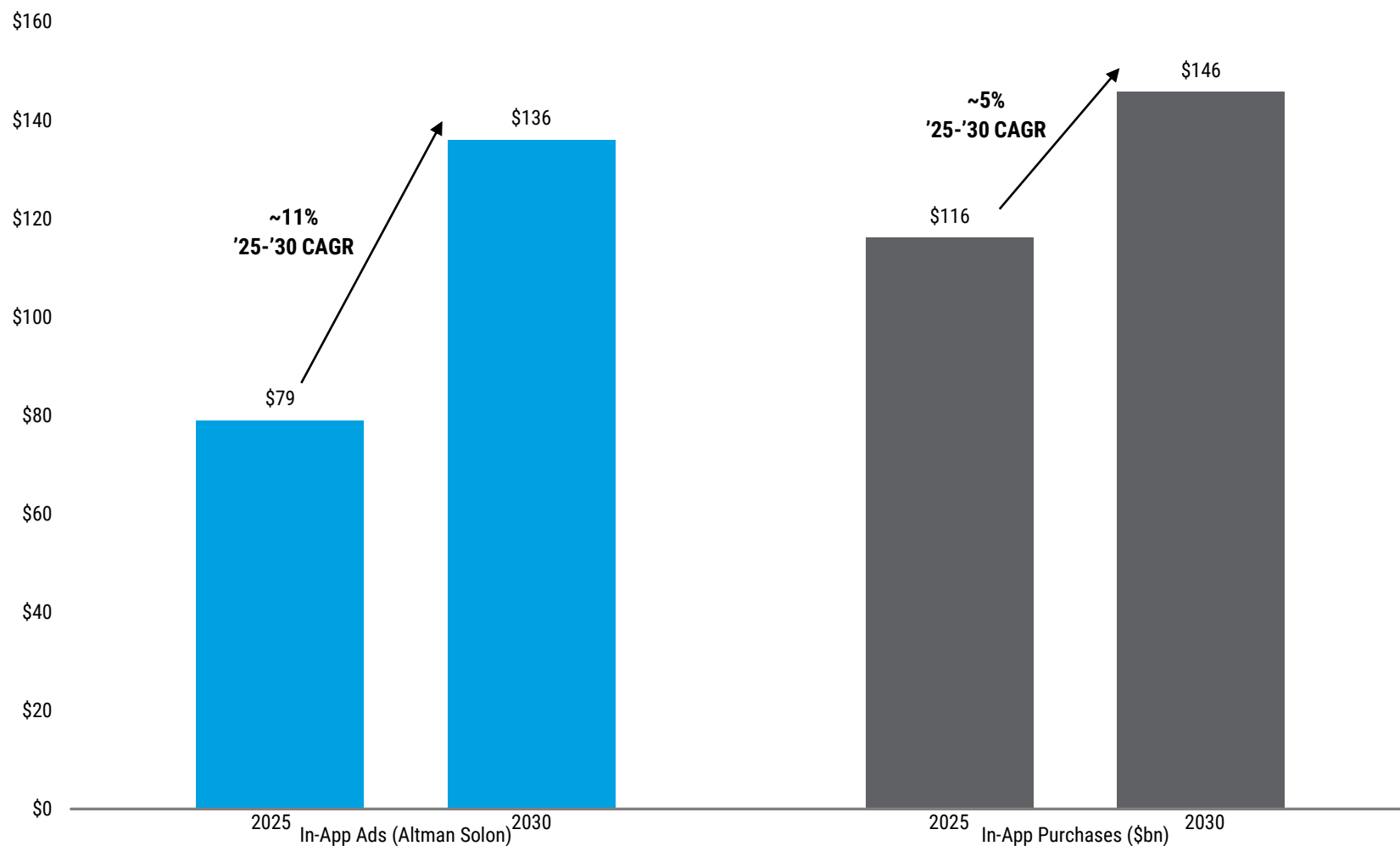
*According to Altman Solon

**According to eMarketer

Source: Company Data, Morgan Stanley Research, Altman Solon, eMarketer

Note: 2029 Number for Global In-App Ad Spend Found by Using 2025-2030 CAGR

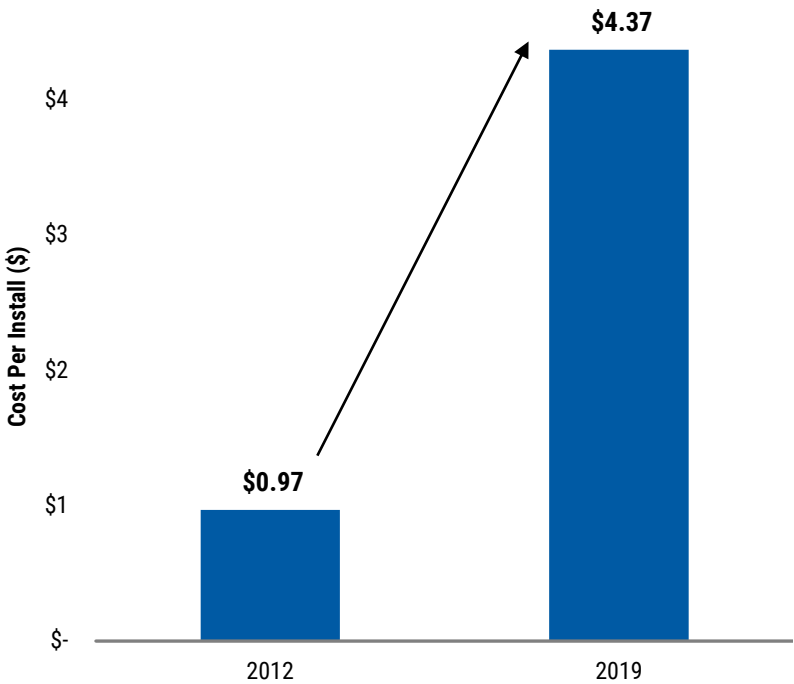
We Expect In-App Advertising to Grow at an 11% CAGR through '30, Materially Faster than Growth on In-Game Purchases at a 5% CAGR...



...But This Has Strong Precedent, as User Acquisition Costs for Mobile Games
Materially Outgrew Spend per User Pre-COVID...

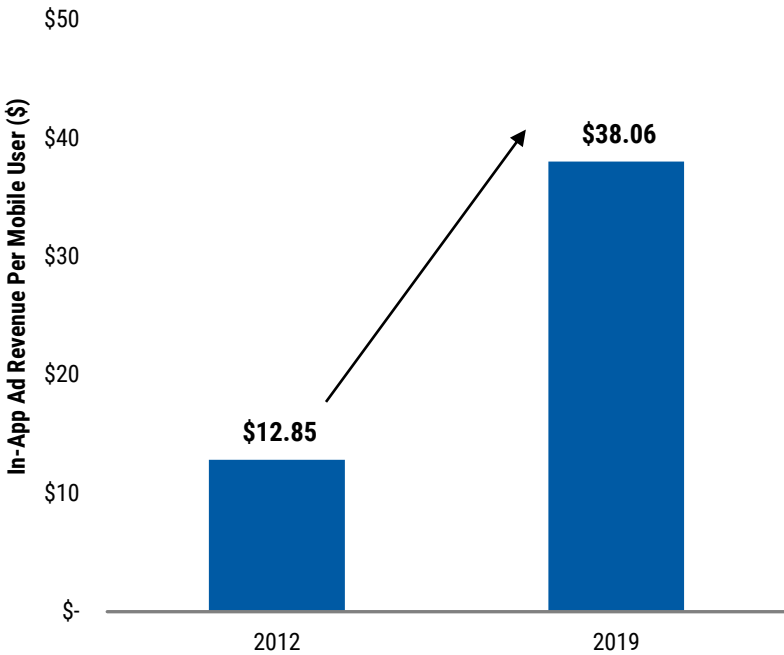
24%

Cost Per Install
2012-2019 CAGR

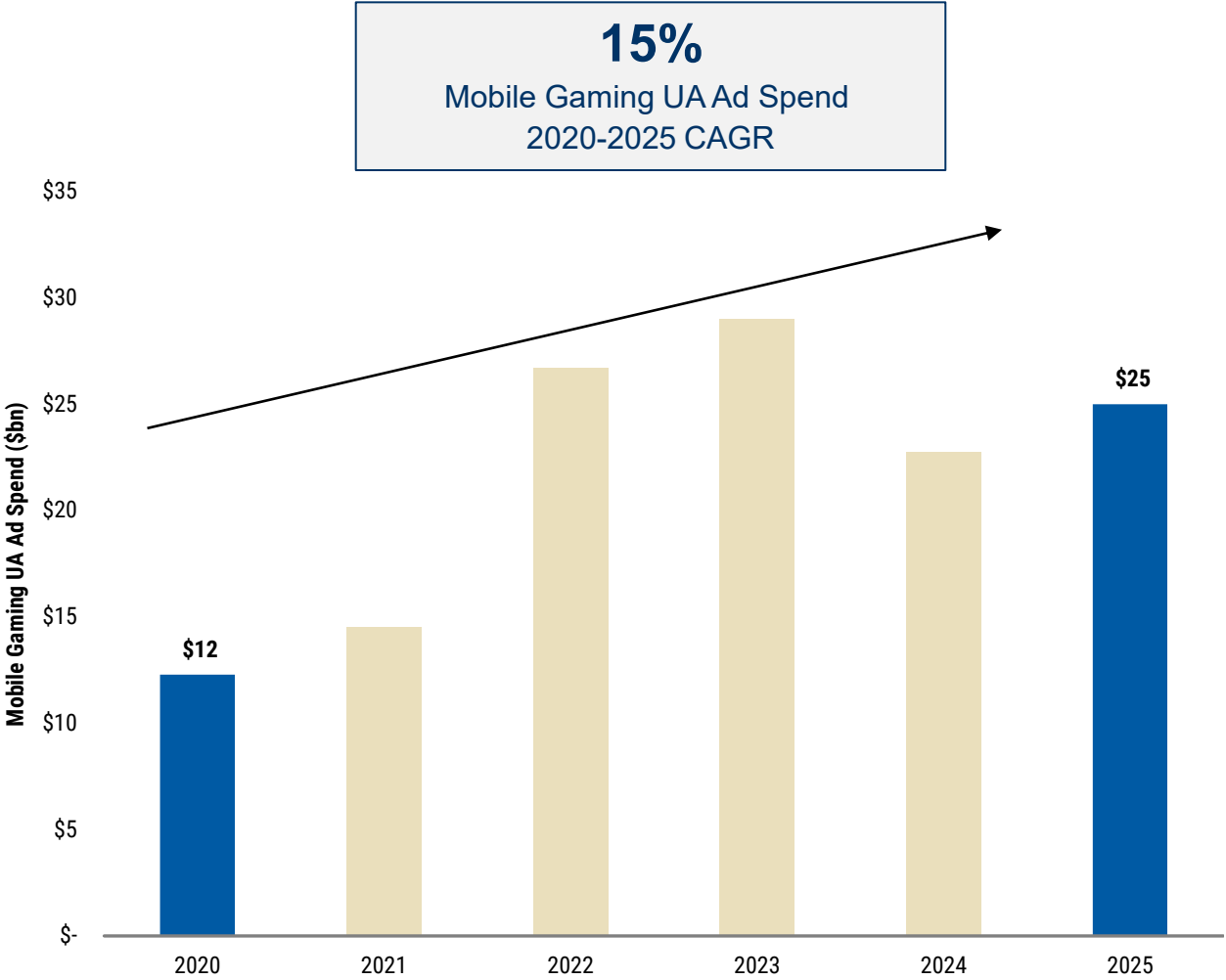


17%

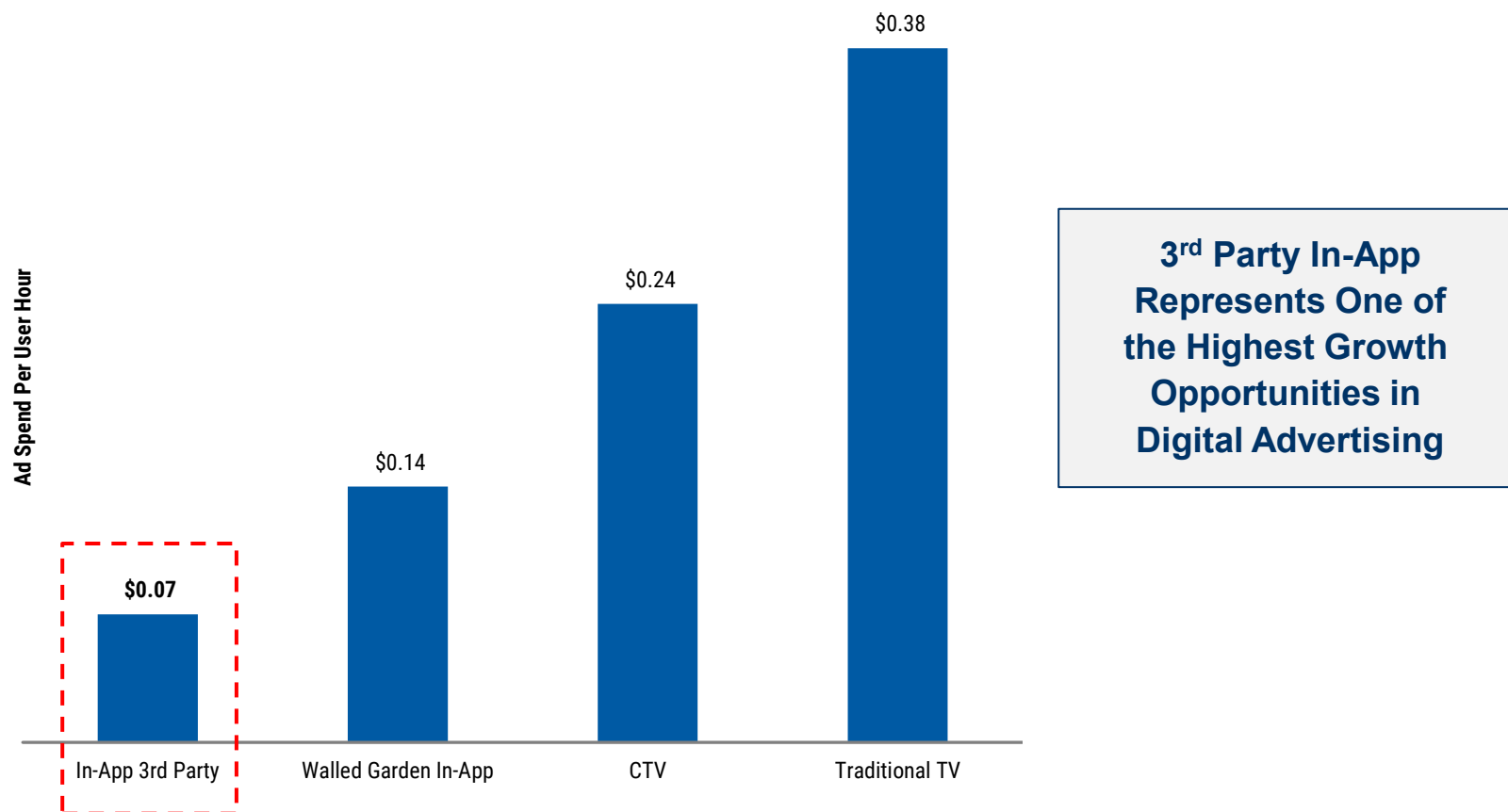
Average Revenue Per Mobile Gamer
2012-2019 CAGR



...and the Trend Has Continued in Recent Years, With Mobile Gaming UA Spend Growing at a 15% '20-'25 CAGR vs. IAP Growing Just 3% Over the Same Period



In-App Ad Spend Arguably Remains Under-Monetized Relative to Other Ad Channels



Who Are Some of the Players in Mobile Ad Tech and What Roles Do They Play?

Company	DSP	SSP	Mediation
 APPLOVIN	✓		✓
 Unity®	✓	✓	✓
 Meta	✓		
 Google	✓	✓	✓
 Mintegral	✓	✓	
 INMOBI™	✓	✓	
 MoLoCo	✓	✓	
 CloudX		✓	✓

What Do Those Roles Mean?

Advertisers

- Companies aiming to acquire new users
 - In this market, typically app-based businesses or those with an app
-

DSP

- Software that helps advertisers buy digital ads programmatically (i.e. in an automated way)
 - Aids advertisers in targeting the right users and selecting creative to achieve their objective by analyzing all available and relevant data, as well as learning from past ad campaigns
-

SSP

- Software that publishers use to make their ad inventory available to many potential buyers (generally DSPs) as possible in an auction format
 - Typically applies rules and works towards the goal of maximizing revenue for each impression
-

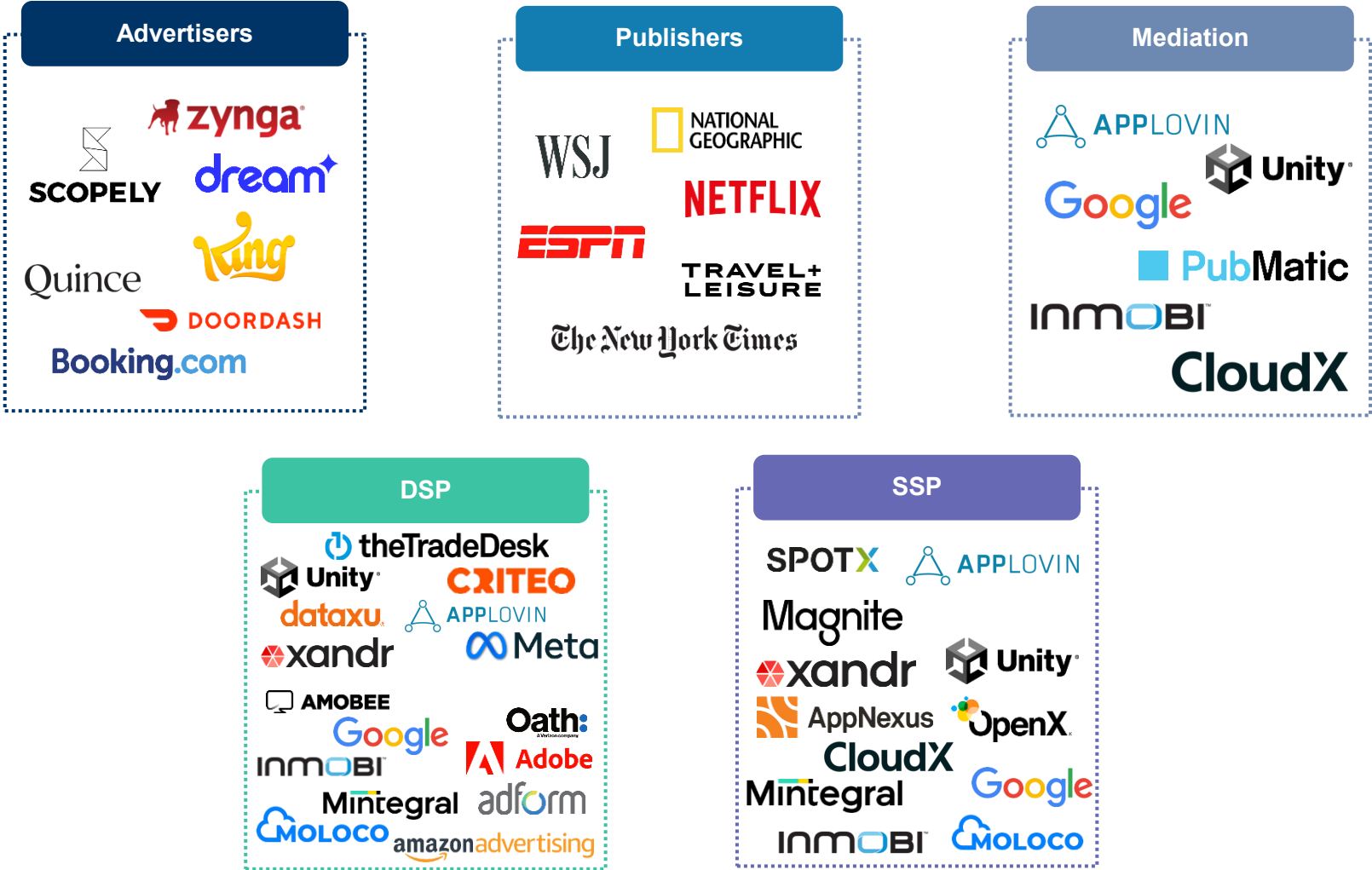
Mediation

- Software that publishers use to connect to multiple SSPs, compare their bids or expected value, and serves the best-paying ad
 - Generally compare bids in real time auction format rather than in series (i.e. waterfall) as in the past
-

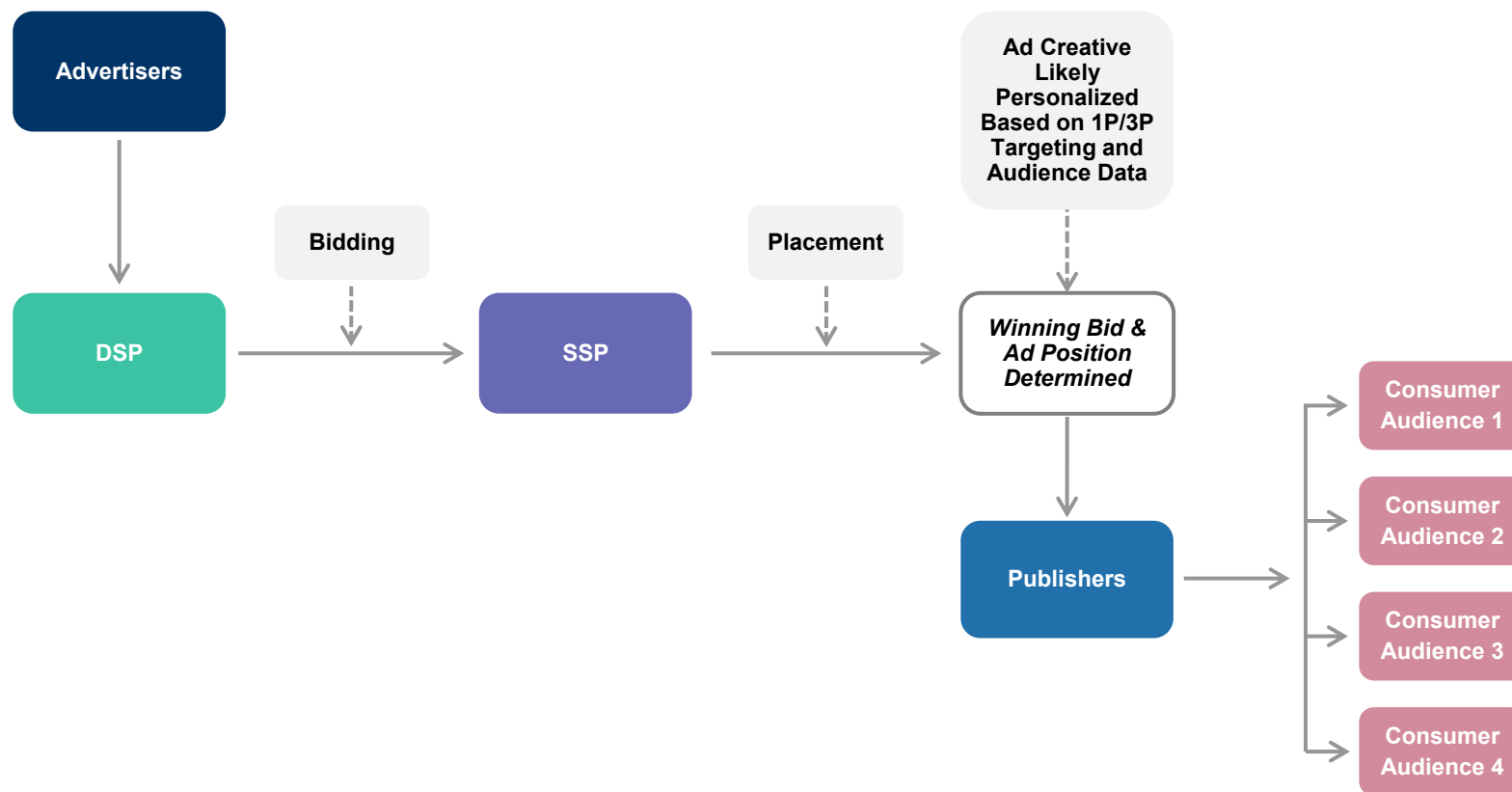
Publishers

- Companies aiming to show ads inside their app, site, or service for profit
 - In this market, typically app-based businesses or those with an app
-

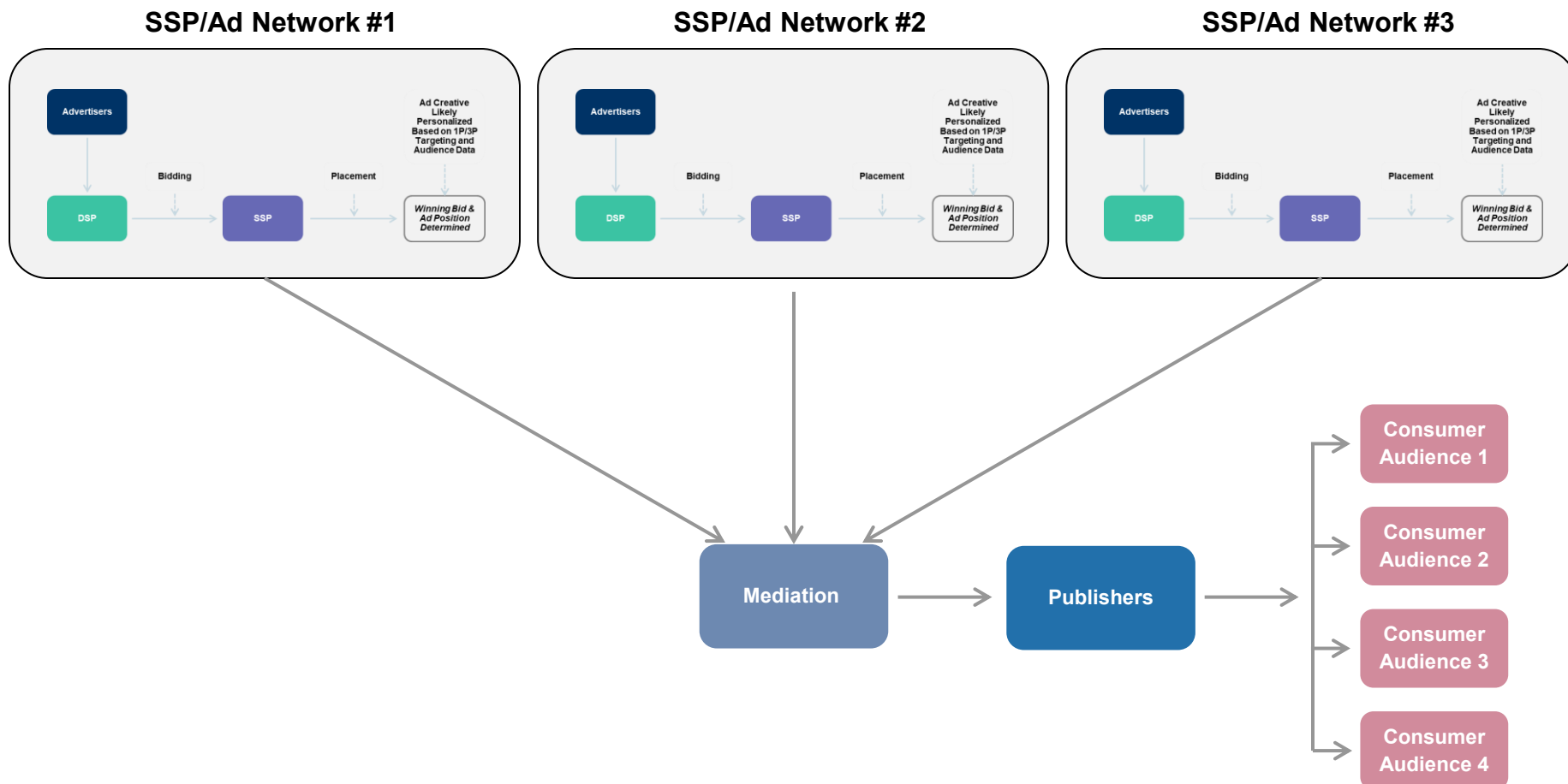
There Are Many Players Across Public/Private Markets, With Many Filling Multiple Roles



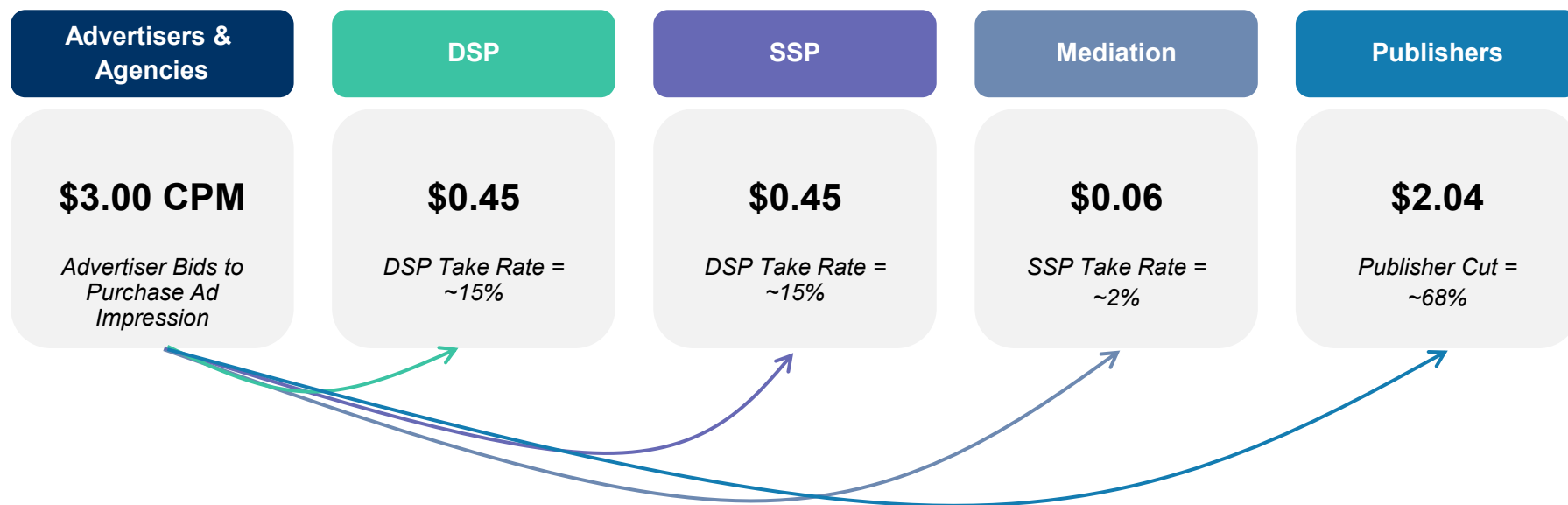
In a Classical Example, Advertisers Execute Campaigns via DSPs, Which Bid into Auctions Run by SSPs on Behalf of Publishers



Mediation Tools Allow Publishers to Work with Multiple SSPs or Ad Networks at Once, With the Goal of Maximizing the Value of Publishers' Ad Inventory



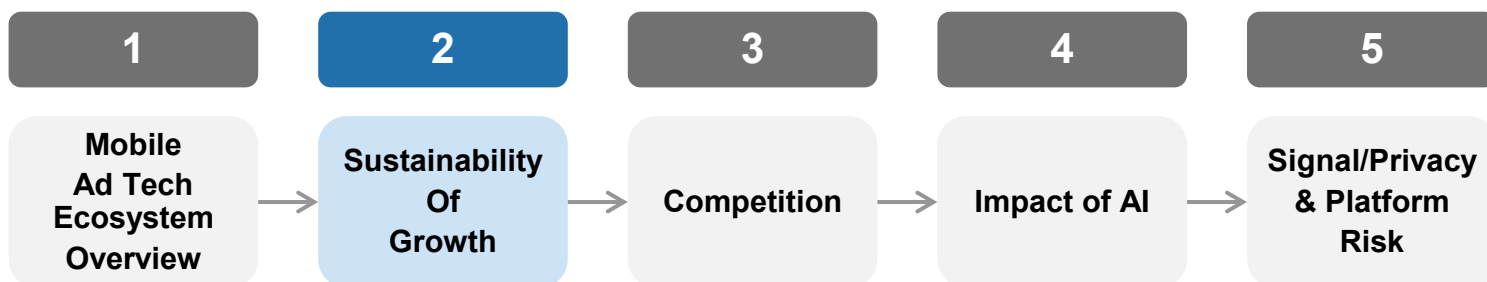
The Majority of the Unit Economics Accrue to Publishers, with DSPs and SSPs Capturing Most of the Remainder



Any Other Potential Fees?

- Ad servers
- Data management platforms (estimated 1-2% but can be included within the DSP take rate)
- Digital media analytics fees are paid separately by the advertiser (i.e. not a percentage of CPM)

Intro to Mobile Ad Tech



Introduction: Sustainability of Growth

What Is The Debate?

- Broadly speaking, there is debate around how much room there is for further growth in mobile advertising, including improvement in existing unit economics or performance, expansion into new verticals, and exposure to macroeconomic pressure.
- Specifically, this includes discussion of how high advertisers can raise take rates and conversion rates, which directly impacts potential growth in revenue for ad tech players.
- Debate has also centered on how scalable ecommerce and other verticals outside of in-game advertising are.

What's Changing?

- Recent initiatives from ad platforms to expand beyond core in-game advertising and into web and e-commerce have driven debate around the durability and scale of these new verticals
- Potential entry of larger platforms into in-game advertising has raised questions regarding the sustainability of conversion rates and take-rate expansion for incumbent players. Separately, there is investor focus on exposure to broader macroeconomic headwinds and potential cyclicalities.

Why Does It Matter?

- Take rates and conversion rates are key drivers of monetization efficiency for platforms and directly impact revenue. Improvement in these metrics signals stronger advertiser ROI and platform pricing power, while any degradation may indicate rising competition or weakening demand quality.
- Expansion beyond in-game advertising could meaningfully increase TAM, supporting higher long-term revenue and margin growth

Who Is Primarily Impacted?

- **Potential Beneficiaries:** APP, U, and other players that can meaningfully scale into new verticals while improving conversion rates.
- **Potentially At Risk:** Existing ad platforms in new verticals that fail to outperform other players on conversion rates or advertiser-specific performance metrics (ROAS, etc.).

Higher Conversion Rates Allow Ad Tech Companies to Increase Their Take Rates and Net Revenue...

What Are Conversion Rates and Take Rates? Why Do They Matter?

- **Click Through Rates:** the portion of users that click on an ad after viewing it
 - **Conversion Rates:** the portion of ad impressions (clicks) that result in a desired outcome (app install, purchase, etc.)
- **“Take Rates”:** The share of advertiser spend captured by the ad platform, contributing to net revenue for the platform
- **Media Spend:** Spend by the ad platform to win placements and impressions to result in one conversion

Total Impressions:
Users who view the ad

x Click Through Rate

Total Engagement:
Users who interact with the ad

x Conversion Rate

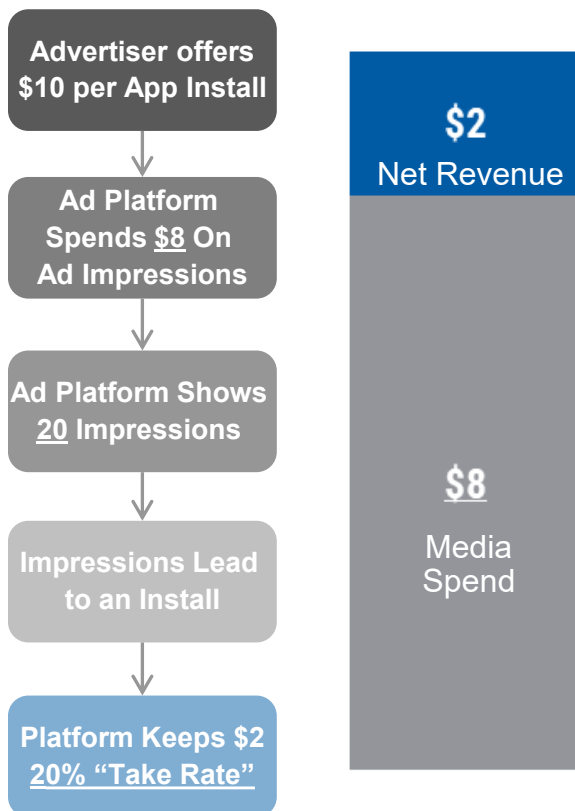
Conversions
*Users Who Complete
The Desired
Outcome*

- **Higher conversion rates result in better campaign efficiency, increasing the value of each impression.** If conversion rates improve, the ad platform needs fewer impressions to deliver the same number of installs.
- **Improved conversion lowers effective CPI for the ad platform** by lowering the number of impressions needed to result in one conversion. **If advertiser spend remains stable, the platform keeps a larger spread between spend and inventory cost.**

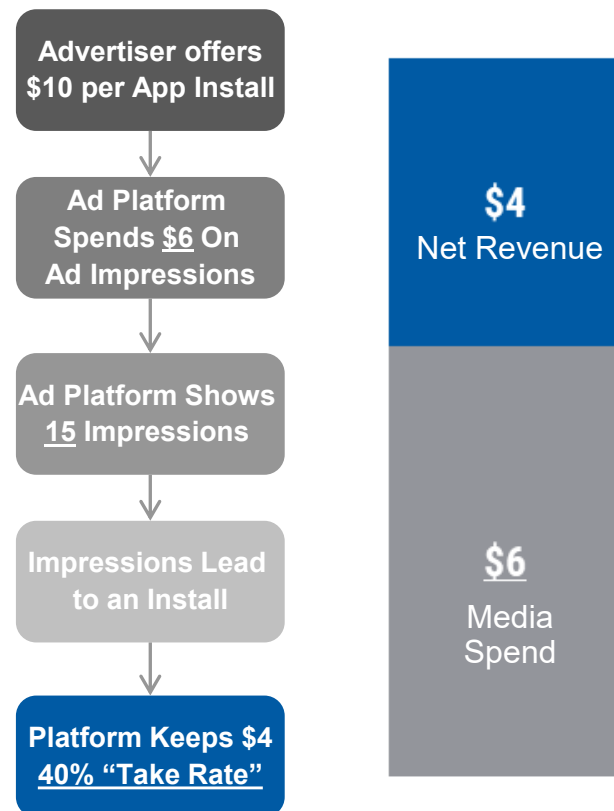
...Presenting a Growth Opportunity that Is Separate from Growth in Total Ad Dollars Spend by Advertisers

Improved Conversion Rates Can Improve The Effective “Take Rate” Of An Ad Platform, Increasing Net Revenue Without Relying on Change in Advertiser Spend. With higher conversion rates, Ad platforms need to buy less impressions to results in one conversion.

If It Takes 20 Ad Impressions To Generate One App Install Today.....



...An Improvement To 15 Impressions Per Install Would Lead "Take Rate" To Double



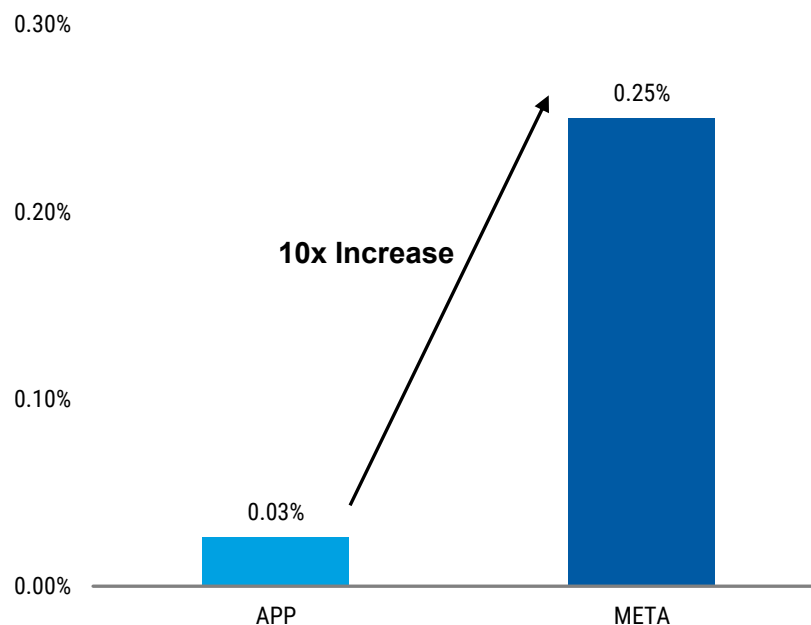
We Estimate Leading Platforms Like META Enjoy “True” Conversion Rates ~10x Above APP’s, Suggesting Significant Headroom Even If APP’s Rates Never Reach META’s

META Conversion Rates are ~10x Those of APPs, Representing Significant Potential Upside to APP Revenue If Conversion Rates and Platform Performance Continue to Improve

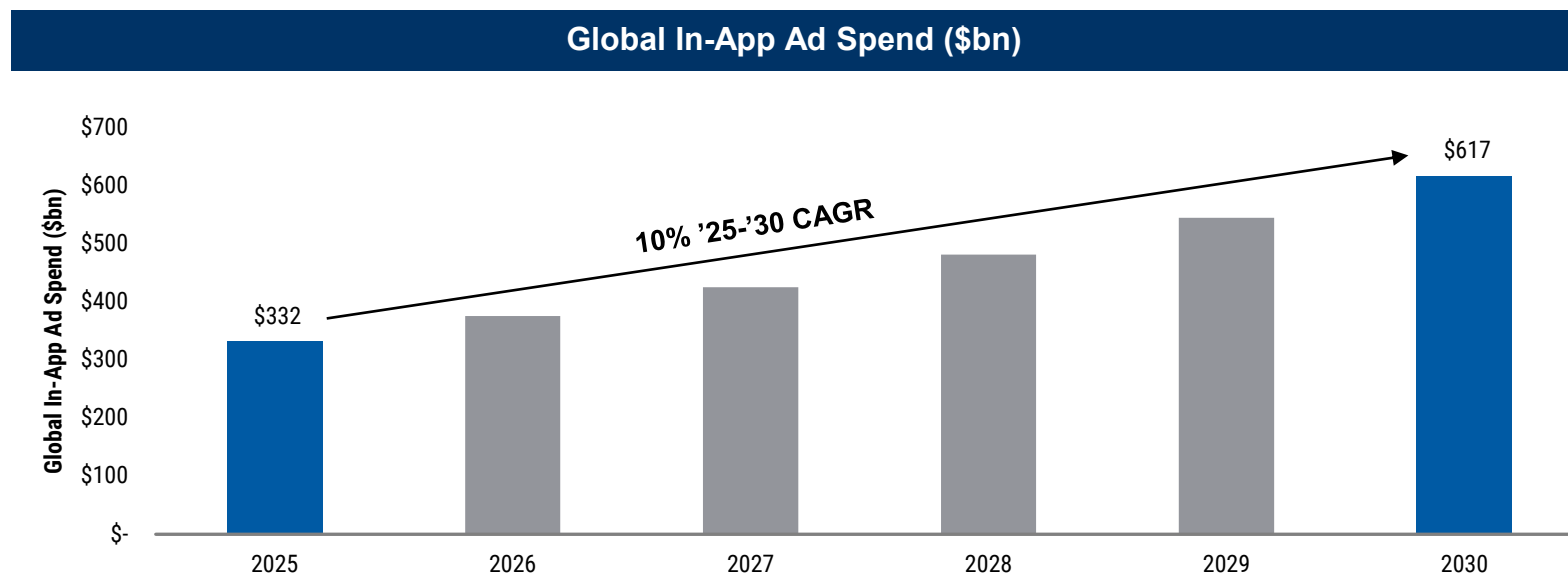
APP	
% of Ads that Generate an App Install	1.3%
x In-Game Paying Ratio	2.0%
= "True" Conversion Rate	0.026%

META	
Click-Through Rate	2.5%
x Conversion Rate	10.0%
= "True" Conversion Rate	0.25%

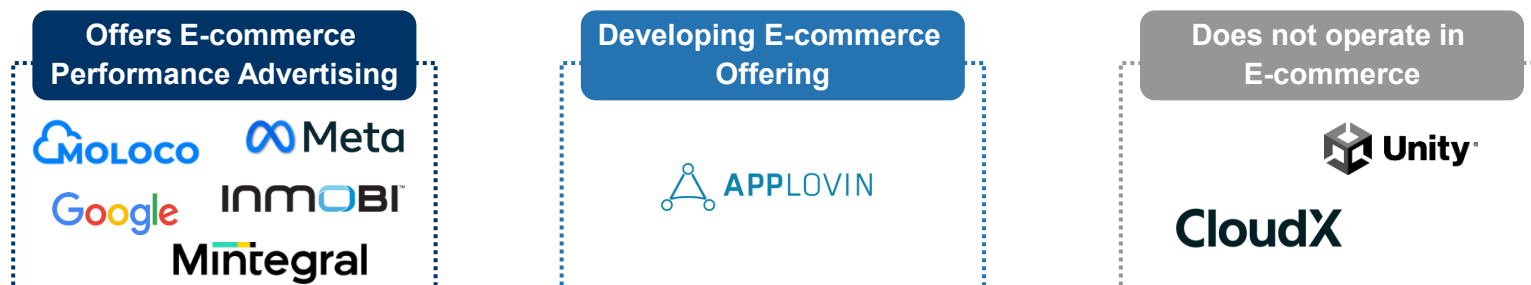
META "True" Conversion Rate	0.25%
/ APP "True" Conversion Rate	0.026%
= Headroom For Expansion	10x



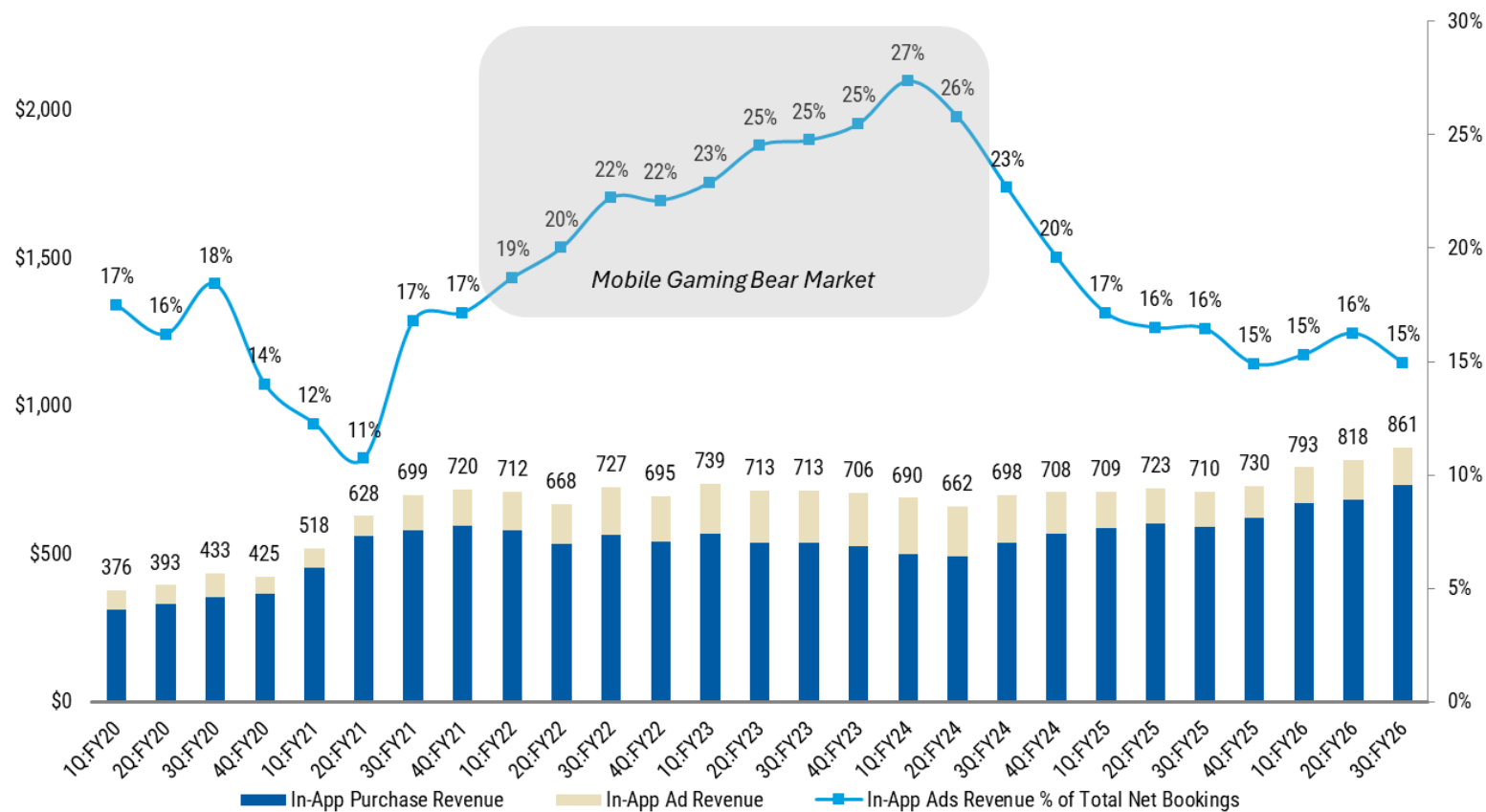
The Serviceable Market for Ad Networks Like APP and U Is Ultimately a Subset of the Estimated \$332bn TAM for In-App Advertising, Which Includes the Walled Gardens



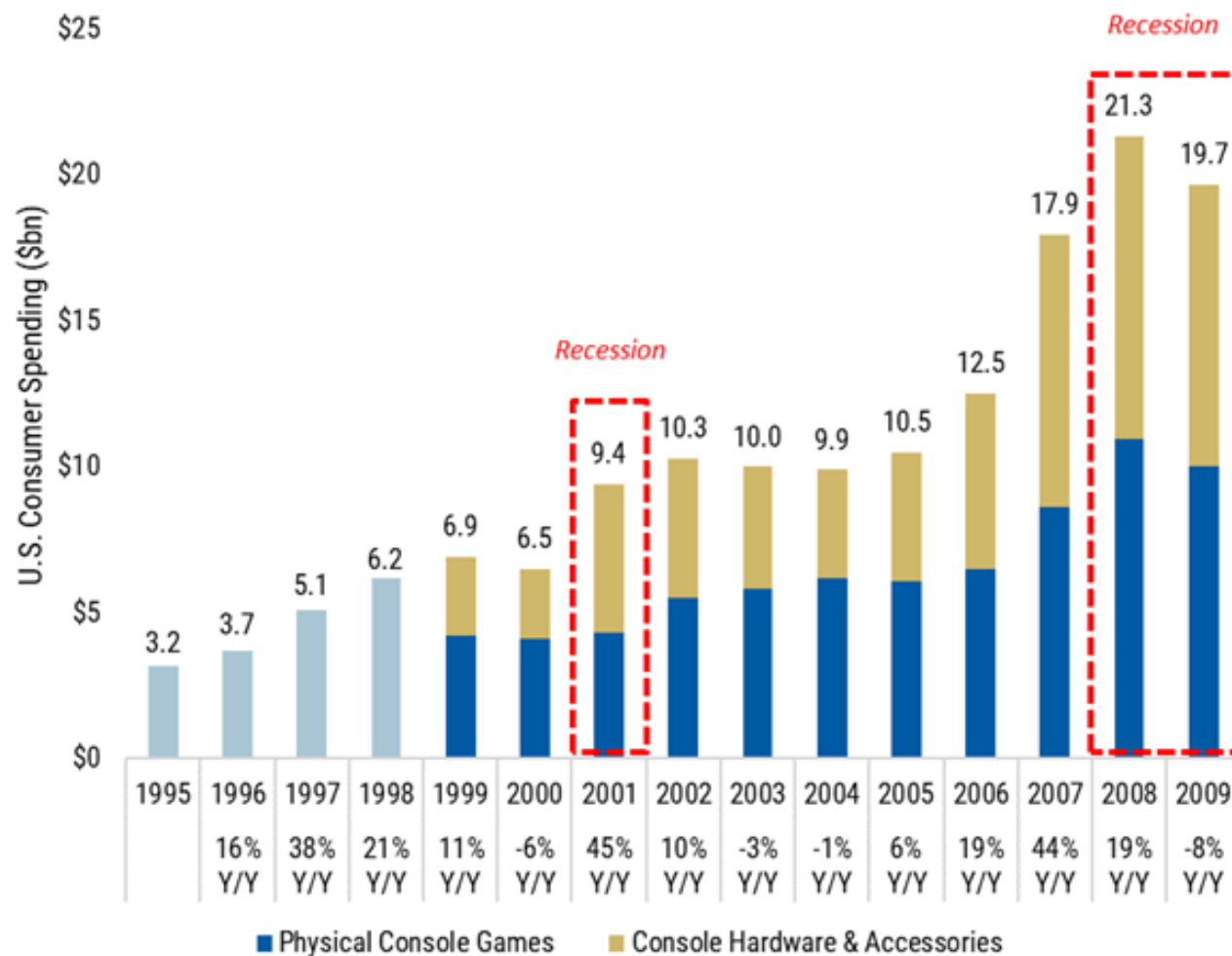
Most Ad Platforms Either Currently Operate or are Building Offerings in Web/E-commerce Within Their Existing Mobile Performance Marketing Businesses



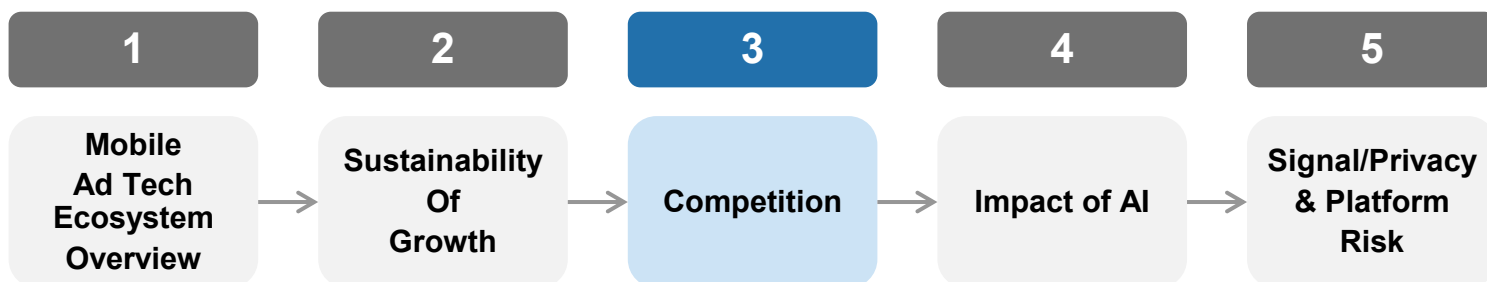
In Periods of Macro/Consumer Weakness, We Have Seen Game Companies Lean Into Monetizing via Ads, Which We See As an Important Cushion for the Ads Ecosystem...



... While Spend on Interactive Entertainment Has Also Generally Performed Well Through Periods of Consumer Weakness



Intro to Mobile Ad Tech



Introduction: Competition

What Is The Debate?

- Large players like GOOGL and META compete in similar ecosystems as smaller, more specialized incumbents like APP, U, Moloco, and others. Historically APP and U primarily operate in in-game advertising, which is separate from the verticals that GOOGL and META operate in.
- Debate has centered on the impact to APP/U if larger companies like META make a larger push in the in-game performance advertising space.
- We believe incumbent platforms already have amassed data and platform models that can effectively place and monetize ads, making their positions more defensible to new competition in their specific mobile advertising verticals.

What's Changing?

- Industry watchers have suggested META is testing its advertising offerings for performance advertising, which would directly compete with in-game mobile advertisers, particularly APP and U.
- New entrants, including mediation platform CloudX, have sparked investor concern around the competitive positioning of incumbent performance advertising platforms.

Why Does It Matter?

- **For incumbents**, new entrants that can outperform through scale or performance could potentially disrupt their market share and competitive positioning in mobile performance advertising.
- **For new entrants**, gaining share in this market unlocks incremental revenue and EBITDA if they can effectively outperform and replace incumbent platforms.

Who Is Primarily Impacted?

- **Potential Beneficiaries:** Platforms and ad tech providers that can deliver the highest return and most efficient ad placement will benefit from increased ad dollars.
- **Potentially At Risk:** Platforms that are unable to stay competitive with industry leaders in terms of ROAS or other ROI metrics for advertising risk losing customers and ad spend on their platforms.

While CloudX Could Theoretically Compete with APP's Mediation Platform, META and APP Operate In Distinct Parts Of The Ecosystem, Limiting Direct Competition

IDFA:

- A Unique, Device-Level Identifier on iOS
- Allows Advertisers to Track User Activity Across Apps For Targeting & Attribution

- **Opt-in:** User Explicitly Accepts Tracking Via ATT, Allowing Advertisers To Access IDFA For Targeting
- **Opt-out:** User Denies Tracking, Advertisers Must Rely On Probabilistic Targeting



DSP

- Bids primarily on IDFA opt-in traffic
- Recent reports suggest that META was testing new products for in-game advertising
- But evidence appears limited that META is focused heavily on in-game ads, and new products appear focused on IDFA opt-in traffic



DSP

Mediation

- Bids primarily on IDFA opt-out traffic
- Scaled DSP and mediation platform in in-game advertising
- Considerable competitive moats from MAX mediation platform.

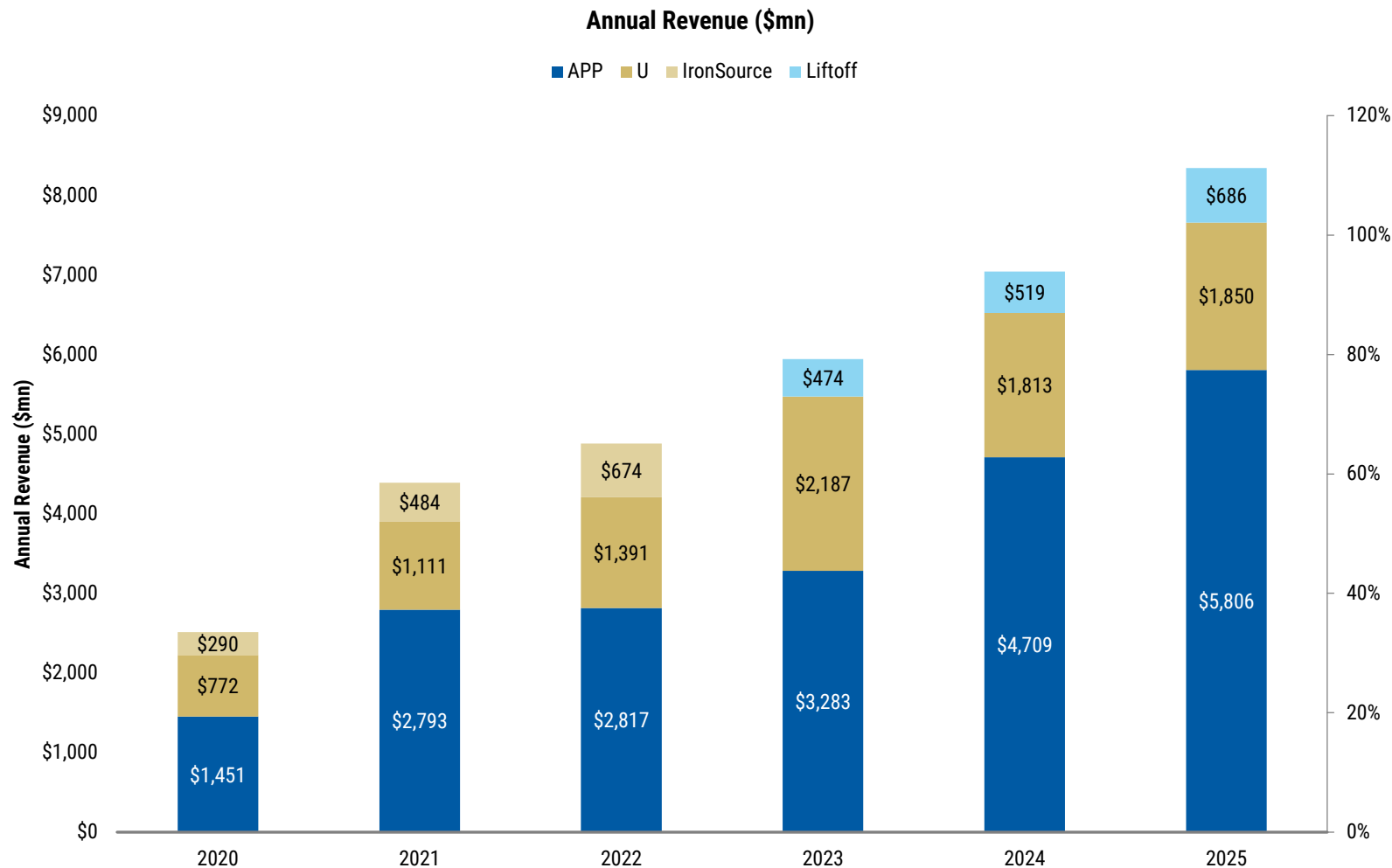


SSP

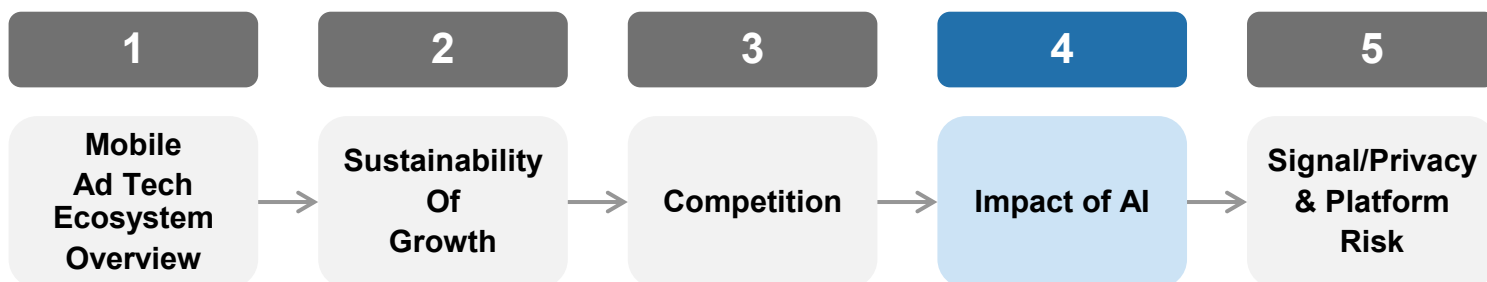
Mediation

- New Entrant, reported to have launched a new tool with demand from META
- Operates as an independent platform, can solicit demand from any source
- Faces significant scale and demand deficits when compared to APP's MAX

Industry Growth Can Support Multiple Platforms as Ad Spend Can Increase Simultaneously Across the Ecosystem



Intro to Mobile Ad Tech



Introduction: Impact of AI

What Is The Debate?

- Recent developments in AI could automate the game development process. While this continues the broader trend of lower barriers to entry for game development, it also creates potential risk that new tools could eventually replace existing platforms used to build games and acquire users.
- This has sparked debate around the impact that new technology will have on discovery and ad tech platforms, with expectations ranging from significant disruption to demand tailwinds.

What's Changing?

- As AI tools are increasingly integrated into game development processes, there is debate about how consumers will interact with mobile games in the future, and if AI models will disrupt current existing user acquisition and monetization platforms.
- In our view, AI tools will lead to the proliferation of content and lower barriers to entry for game development, meaning that there will be significantly more games and content for consumers to interact with. This would increase the value and demand efficient user acquisition and monetization platforms, not disrupt them.

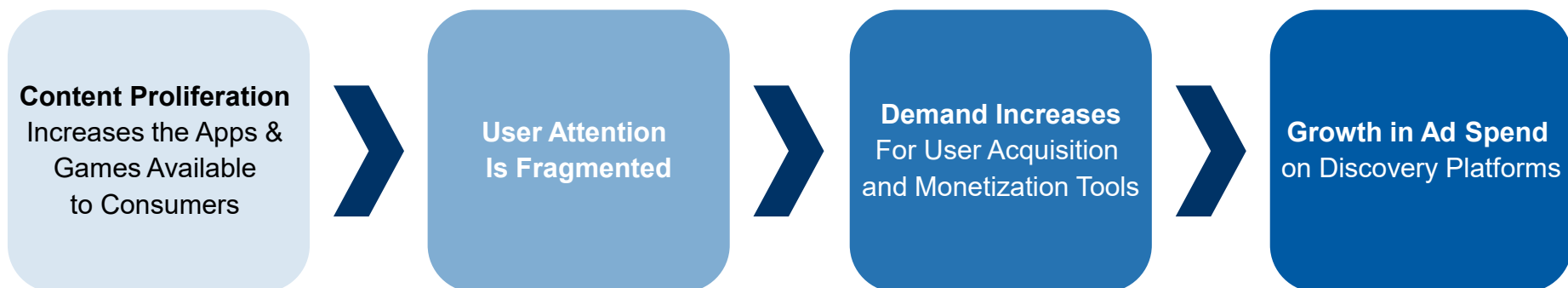
Why Does It Matter?

- In a scenario where AI lowers the barriers to entry of game development, incumbent ad tech/discovery platforms will likely have higher demand and value for customers trying to acquire and monetize new users.
- In a scenario where video games are distributed on entirely new platforms, incumbent discovery players may struggle to integrate their models into these new game platforms, which poses a potential deplatforming risk.

Who Is Primarily Impacted?

- **Potential Beneficiaries:** In our view, ad tech is positioned to benefit incrementally from the lower barriers to entry for game development with AI. A proliferation of content would increase the value of efficient advertising platforms.
- **Potentially At Risk:** Ad providers that are not integrated into new game distribution platforms which are dominated by internal advertising offerings.

Disruption to the Game Industry Is Possible, But Is Likely to Boost Demand For Discovery and Monetization Tools



Even As New Platforms Redefine UGC Creation and Interaction, 3rd Party Advertising Will Likely Remain Important, Insulating Ad Platforms from Disruption

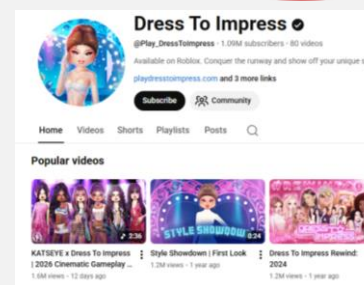
77%

Of Creators Use Social Media As A Top Audience Channel

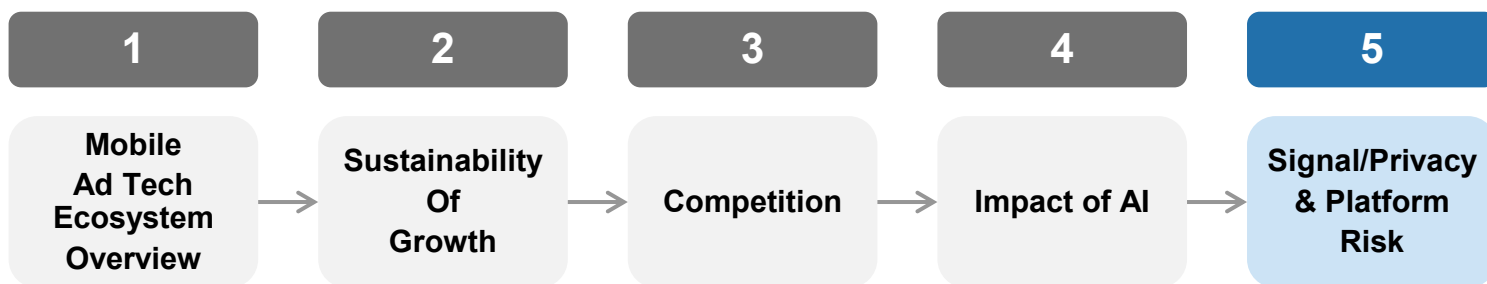
60%

Of Creators Rely On Social Platforms to Showcase Work

Content Native to UGC Platforms Is Commonly Promoted or Shared Using Other Sites. Roblox Experience 'Dress to Impress' Has been promoted on platforms YouTube and Instagram, showcasing how UGC content is promoted outside of the site it was created on.

UGC-Native Platform**ROBLOX****3rd – Party Platform Promotion****You Tube**

Intro to Mobile Ad Tech



Introduction: Signal, Platform, & Privacy Risk

What Is The Debate?

- 3rd party ad tech providers rely on data signals from platforms like GOOGL and AAPL, who control the operating systems and grant access to user/app data.
- Ongoing privacy changes have reduced the quality of user data available to ad platforms, which can potentially weaken the targeting and performance for ad providers.
- At the same time, total deplatforming from GOOGL or AAPL could completely remove access to this data, affecting an advertising platform's ability to function.
- There is also ongoing legal and regulatory scrutiny around data privacy and user tracking practices for advertising platforms.

What's Changing?

- Privacy changes from the Apple App Store and Google Play Store have reduced the granularity of user data available to ad platforms, which could potentially weaken their performance.
- Ongoing regulatory scrutiny and rumored investigations into data collection practices have sparked debate on the potential for further restriction or rules.

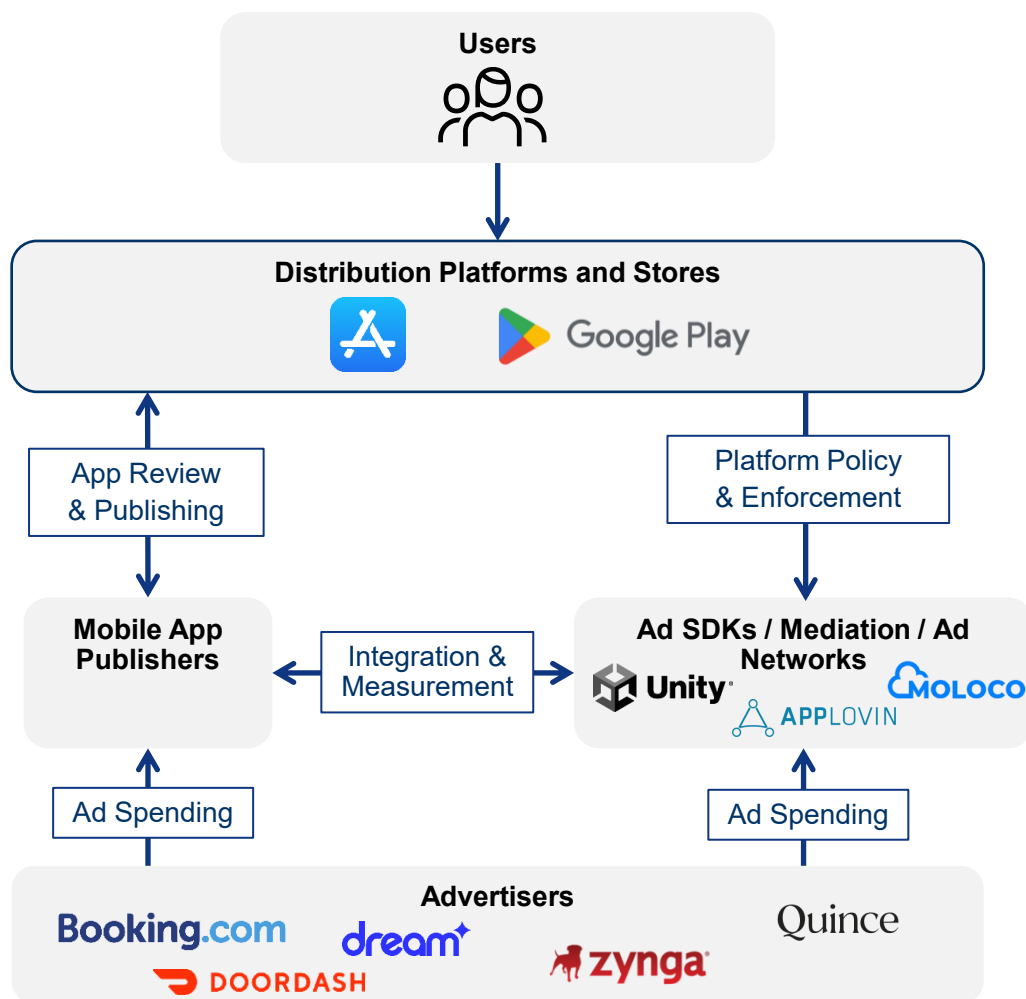
Why Does It Matter?

- Weaker signal or data provided by platforms could lower the performance of existing advertising technology, while complete deplatforming could revoke access to the data vital for ad tech businesses to operate.
- Increased regulation around data collection practices could negatively impact the performance of existing platforms, resulting in lower advertising spend and revenue for the platforms.

Who Is Primarily Impacted?

- **Potential Beneficiaries:** Limited, but includes platform-native ad solutions from AAPL or GOOGL
- **Potentially At Risk:** Incumbent mobile ad tech players that use consumer data to target and measure ads, but also includes any ad business that does not control a mobile OS (i.e. other than AAPL/GOOGL)

Mobile Ad Platforms Rely on Access to Users, SDK Integration, and Measurement Signals That are Controlled by Distribution Platforms Including Apple and Google



Platforms control the mobile distribution layer. Ad networks depend on approved SDK access, measurement signals, and publisher app inventory to monetize advertiser demand.

Since the platforms have control over setting and enforcing rules or privacy policies, they could potentially revoke access or restrict data that ad platforms currently rely on to operate.

Policy changes or deplatforming can impair targeting, attribution, and monetization performance of the platforms.

Previous Data Access Removal on Platforms has Materially Impacted Advertiser Spending and Ad Platform Revenue

What is IDFA?

IDFA (Identifiers For Advertisers): A unique, device-level identifier on iOS that allows advertisers to track user activity across apps.

How Did ATT Removal Impact Ad Platform Performance?

While ATT removed user-level tracking, Advertisers rebuilt targeting and attribution with machine learning.

Pre-ATT

- IDFA Available By Default For Most Users
- Users Had To Intentionally Opt-out of Tracking
- Advertisers Had Deterministic Tracking Across Apps within iOS

Strong Tracking, Attribution And Measurement



April 2021

- Apple Launches App Tracking Transparency (ATT)
- Apps Must Now Ask Users To Opt-in To Tracking
- Most Users Opted-Out of Tracking, Removing Attribution/Measurement for Ad Platforms

Majority Of Users Opt-out → Sharp Signal Loss

Post-ATT

- AAPL Limits Access to IDFA Through ATT
- GOOGL Prohibits Fingerprinting Devices for Tracking
- Reduced Targeting Precision Caused a Shift to Probabilistic Attribution & Contextual Targeting

Move to Reliance on 1st Party Data & ML Optimization



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(as of May 31, 2026)

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For disclosure purposes only (in accordance with FINRA requirements), we include the category headings of Buy, Hold, and Sell alongside our ratings of Overweight, Equal-weight, Not-Rated and Underweight. Morgan Stanley does not assign ratings of Buy, Hold or Sell to the stocks we cover. Overweight, Equal-weight, Not-Rated and Underweight are not the equivalent of buy, hold, and sell but represent recommended relative weightings (see definitions below). To satisfy regulatory requirements, we correspond Overweight, our most positive stock rating, with a buy recommendation; we correspond Equal-weight and Not-Rated to hold and Underweight to sell recommendations, respectively.

Disclosure Section (cont.)

Stock Rating Category	Coverage Universe		Investment Banking Clients (IBC)			Other Material Investment Services Clients (MISC)	
	Count	% of Total	Count	% of Total IBC	% of Rating Category	Count	% of Total Other MISC
Overweight/Buy	1542	42%	465	51%	30%	707	43%
Equal-weight/Hold	1571	43%	369	40%	23%	723	44%
Not-Rated/Hold	3	0%	0	0%	0%	1	0%
Underweight/Sell	551	15%	86	9%	16%	201	12%
Total	3,667		920			1632	

Data include common stock and ADRs currently assigned ratings. Investment Banking Clients are companies from whom Morgan Stanley received investment banking compensation in the last 12 months. Due to rounding off of decimals, the percentages provided in the "% of total" column may not add up to exactly 100 percent.

Analyst Stock Ratings

Overweight (O). The stock's total return is expected to exceed the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Equal-weight (E). The stock's total return is expected to be in line with the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Not-Rated (NR). Currently the analyst does not have adequate conviction about the stock's total return relative to the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Underweight (U). The stock's total return is expected to be below the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Unless otherwise specified, the time frame for price targets included in Morgan Stanley Research is 12 to 18 months.

Analyst Industry Views

Attractive (A): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be attractive vs. the relevant broad market benchmark, as indicated below.

In-Line (I): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be in line with the relevant broad market benchmark, as indicated below.

Cautious (C): The analyst views the performance of his or her industry coverage universe over the next 12-18 months with caution vs. the relevant broad market benchmark, as indicated below.

Benchmarks for each region are as follows: North America - S&P 500; Latin America - relevant MSCI country index or MSCI Latin America Index; Europe - MSCI Europe; Japan - TOPIX; Asia - relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

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INDUSTRY COVERAGE: Internet

COMPANY (TICKER)	RATING (AS OF)	PRICE* (06/04/2026)
Brian Nowak, CFA		
Airbnb Inc. (ABNB.O)	U (12/06/2022)	\$133.72
Alphabet Inc. (GOOGL.O)	O (08/11/2015)	\$372.19
Amazon.com Inc. (AMZN.O)	O (04/24/2015)	\$253.79
Booking Holdings Inc. (BKNG.O)	O (02/23/2026)	\$167.49
DoorDash Inc. (DASH.O)	O (02/21/2024)	\$160.07
Expedia Inc. (EXPE.O)	E (01/09/2019)	\$227.18
Instacart (CART.O)	E (01/29/2024)	\$41.48
Lyft Inc. (LYFT.O)	E (10/24/2019)	\$14.12
Meta Platforms Inc. (META.O)	O (03/20/2023)	\$627.57
Pinterest Inc. (PINS.N)	O (07/20/2025)	\$21.59
Reddit Inc. (RDDT.N)	O (12/08/2024)	\$183.91
Snap Inc. (SNAP.N)	E (07/22/2024)	\$6.07
Uber Technologies Inc. (UBER.N)	O (06/04/2019)	\$72.21
Matthew Cost		
AppLovin Corp (APPO)	O (04/10/2025)	\$558.87
Compass, Inc. (COMP.N)	E (01/12/2026)	\$7.88
Criteo SA (CRTO.O)	E (01/26/2016)	\$17.29
DoubleVerify Holdings Inc (DV.N)	E (06/25/2024)	\$10.57
Electronic Arts Inc (EA.O)	E (08/04/2021)	\$203.40
MNTN Inc. (MNTN.N)	E (06/16/2025)	\$9.69
Opendoor Technologies Inc (OPEN.O)	E (07/24/2023)	\$4.95
Playtika Holding Corp (PLTK.O)	E (11/27/2022)	\$3.15
Roblox Corporation (RBLX.N)	O (11/04/2024)	\$43.35
Shutterstock Inc (SSTK.N)	E (07/28/2022)	\$13.19
Take-Two Interactive Software (TTWO.O)	O (02/01/2018)	\$216.65
Trade Desk Inc (TTD.O)	E (09/10/2025)	\$21.03
Unity Software Inc (U.N)	O (09/02/2024)	\$30.03
Webtoon Entertainment Inc (WBTN.O)	E (07/22/2024)	\$11.66
Yelp Inc (YELP.N)	U (01/10/2019)	\$23.63
Zillow Group Inc (Z.O)	E (04/18/2018)	\$35.92
Nathan Feather		
Bumble Inc. (BMBL.O)	E (03/08/2021)	\$2.99
Chewy Inc (CHWY.N)	O (10/31/2023)	\$20.82
Duolingo Inc (DUOL.O)	E (02/27/2026)	\$109.15
eBay Inc (EBAY.O)	O (04/18/2024)	\$109.15
Etsy Inc (ETSY.N)	E (07/20/2025)	\$67.05
FIGS, Inc. (FIGS.N)	E (02/29/2024)	\$11.70
Grindr Inc. (GRND.N)	E (02/24/2026)	\$10.70
Match Group Inc (MTCH.O)	E (04/18/2024)	\$34.76
Peloton Interactive, Inc. (PTON.O)	E (03/14/2022)	\$6.17
Revolve Group Inc (RVLV.N)	E (10/20/2024)	\$19.33
WW International Inc (WW.O)	E (08/01/2025)	\$17.27

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* Historical prices are not split adjusted.

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